



Individual Documentation Of Disease Detection_AI-Chatbot

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1) Problem Description

Accurate disease prediction based on patient symptoms is a challenging task in the healthcare sector. Many diseases share common symptoms, making manual diagnosis difficult and time-consuming. Early disease detection can assist patients in obtaining timely medical consultation and treatment.

The objective of this project is to develop an intelligent Disease Detection AI Chatbot capable of predicting diseases from user-provided symptoms. The system uses machine learning techniques to learn symptom-disease relationships from historical medical datasets. After training, the model can identify the most probable disease and provide relevant information and precautionary measures.

The chatbot serves as a preliminary diagnostic support system and helps bridge the gap between symptom appearance and professional medical consultation.

2) Methodology

Our project follows a Supervised Machine Learning approach.

- **Model Selection:**

XGBoost (Extreme Gradient Boosting) was selected as the primary classification model. XGBoost is one of the most powerful ensemble learning algorithms and is widely used because of its high predictive performance, regularization capability, and efficient handling of multiclass classification problems.

- **Data Preparation:**

The dataset was loaded from CSV files containing symptom features and disease labels. Duplicate columns were removed, and column names were standardized to ensure consistency during training and prediction.

- **Evaluation Strategy:**

The dataset was divided into training and testing subsets using a 67%-33% split. Model performance was evaluated using Accuracy, Precision, Recall, and F1-Score. In addition, 5-Fold Cross Validation was applied to measure the model's stability and robustness across multiple data partitions.

3) Implementation Details

- **Data Preprocessing:**

- Dataset loaded using Pandas.
- Duplicate symptom columns removed.

- Disease labels encoded using LabelEncoder.
- Features and target labels separated.
- Data split into training and testing sets using train_test_split().

● **Symptom Processing:**

The chatbot includes a symptom extraction mechanism that detects symptoms from natural language user input. Synonym mapping and fuzzy matching techniques are used to recognize symptoms even when users provide alternative spellings or phrases.

Examples:

stomach ache → stomach_pain

loose motion → diarrhea

throat pain → sore_throat

body ache → muscle_pain

● **Backend & Model:**

The backend was implemented in Python. The XGBoost classifier was trained using symptom vectors as input features and disease names as output labels. After training, the model predicts the most probable disease along with confidence scores generated from prediction probabilities.

The chatbot also retrieves:

- Disease descriptions
- Symptom severity information
- Recommended precautions
- from external knowledge files.

4) Results & Analysis

Accuracy : 100.00 %

Precision (Weighted) : 100.00 %

Recall (Weighted) : 100.00 %

F1-Score (Weighted) : 100.00 %

The XGBoost model achieved an Accuracy of 100.00%, indicating that all test instances were classified correctly. The Precision, Recall, and F1-Score also reached 100.00%, demonstrating outstanding classification performance across all disease categories.

These results suggest that the model successfully learned the relationships between symptoms and diseases from the training dataset. The weighted evaluation metrics indicate that the classifier maintained consistent performance across different disease classes without favoring any particular category.

The confusion matrix analysis showed perfect classification on the test dataset, with no observed misclassifications. This indicates that the symptom patterns in the dataset were effectively captured by the XGBoost algorithm.

Furthermore, the ensemble boosting mechanism of XGBoost contributed significantly to the model's predictive power by combining multiple decision trees and minimizing classification errors during training. The model demonstrated excellent generalization capability on the test data and proved highly effective for multiclass disease prediction.

Overall, the XGBoost-based Disease Detection AI Chatbot achieved exceptional performance and can serve as a reliable preliminary disease prediction system based on user-provided symptoms. However, further evaluation on larger and real-world clinical datasets would be beneficial to validate its effectiveness in practical healthcare environments.